



LS Maths Year 8 Programme of Study

Module One	National Curriculum Link	Cambridge Curriculum link	Coursebook Pages
Topics: 1.1 Ordering negative numbers 1.2 Adding and subtracting negative numbers 1.3 Multiplying and dividing negative numbers 1.4 Arithmetic sequences 1.5 Other sequences	- Order integers and decimals. - Use the four operations, including formal written methods, applied to integers. - Generate terms of a sequence from a term-to-term rule. - Generate terms of a sequence from a position-to-term rule. - Recognise geometric sequences. - Model situations or procedures by translating them into algebraic expressions.	7Ni.03 Estimate, multiply and divide integers including where one integer is negative. 7As.01 Understand term-to-term rules, and generate sequences from numerical and spatial patterns (linear and integers). 7As.02 Understand and describe nth term rules algebraically (in the form $n \pm a$, $a \times n$ where a is a whole number). 8As.01 Understand term-to-term rules, and generate sequences from numerical and spatial patterns (including fractions). 8As.02 Understand and describe nth term rules algebraically (in the form $n \pm a$, $a \times n$, or $an \pm b$, where a and b are positive or negative integers (or fractions)). *objectives in brackets are studied at a later stage in Lower Secondary.	Pages 94-107 Pages 1-15
ASSIGNMENT ONE			





Module Two	National Curriculum Link	Cambridge Curriculum link	Coursebook Pages
Topics: 2.1 Addition and subtraction 2.2 Multiplication and division 2.3 Multiplying decimals 2.4 Dividing decimals 2.5 Plotting straight line graphs 2.6 Real life graphs	- Use the four operations including formal written methods. - Use the four operations applied to decimals. - Extend their understanding of the number system. - Work with coordinates in all four quadrants. - Recognise, sketch and produce graphs of linear functions.	7Nf.07 (Estimate and) add and subtract positive and negative numbers with the same or different number of decimal places. 7Nf.08 (Estimate and) multiply and divide decimals by whole numbers. 8Nf.07 (Estimate and) multiply decimals by integers and decimals. 8Nf.08 (Estimate and) divide decimals by numbers with one decimal place. 9Nf.06 (Estimate,) multiply and divide decimals by integers and decimals. 7As.04 Understand that a situation can be represented either in words or as a linear function in two variables (of the form $y = x + c$ or $y = mx$), and move between the two representations. 8As.04 Understand that a situation can be represented either in words or as a linear function in two variables (of the form $y = mx + c$), and move between the two representations. 8As.06 Recognise that equations of the form $y = mx + c$ correspond to straight-line graphs, where m is the gradient and c is the y-intercept (integer values of m) (continued on next page)	Pages 75-89 Pages 21-38





Module Two	National Curriculum Link	Cambridge Curriculum link	Coursebook Pages
		7As.05 Use knowledge of coordinate pairs to construct tables of values and plot the graphs of linear functions, where y is given explicitly in terms of x ($y = x + c$ or $y = mx$). 8As.05 Use knowledge of coordinate pairs to construct tables of values and plot the graphs of linear functions, where y is given explicitly in terms of x ($y = mx + c$). 8As.06 Recognise that equations of the form $y = mx + c$ correspond to straight-line graphs, where m is the gradient and c is the y-intercept (integer values of m) 9As.05 Use knowledge of coordinate pairs to construct tables of values and plot the graphs of linear functions(, including where y is given implicitly in terms of x ($ax + by = c$) and quadratic functions of the form $y = x^2 \pm a$) 8As.07 Read and interpret graphs with more than one component. Explain why they have a specific shape (and the significance of intersections of the graphs.)	
ASSIGNMENT TWO			





Module Three	National Curriculum Link	Cambridge Curriculum link	Coursebook Pages
Topics: 3.1 Angles in a quadrilateral 3.2 Parallel lines 3.3 Polygons 3.4 Protractor use 3.5 Bearings 3.6 Scale drawings 3.7 Constructing triangles	- Use properties of quadrilaterals to calculate angles. - Understand the relationship between parallel lines and alternate and corresponding angles. - Deduce angle sum in any polygon. - Draw and measure angles. - Use scale diagrams. - Draw and measure line segments and angles in geometric figures. - Construct triangles using compasses.	7Gg.11 Derive the property that the sum of the angles in a quadrilateral is 360°, and use this to calculate missing angles. 7Gg.13 Recognise the properties of angles on: - parallel lines and transversals - perpendicular lines - intersecting lines. 8Gg.11 Recognise and describe the properties of angles on parallel and intersecting lines, using geometric vocabulary such as alternate, corresponding and vertically opposite 9Gg.09 Use properties of angles, parallel and intersecting lines, triangles and quadrilaterals to calculate missing angles. 9Gg.07 Derive and use the formula for the sum of the interior angles of any polygon. 9Gg.08 Know that the sum of the exterior angles of any polygon is 360°. 8Gp.01 Understand and use bearings as a measure of direction. 9Gp.01 Use knowledge of bearings and scaling to interpret position on maps and plans. 8Gg.12 Construct triangles, (midpoint and perpendicular bisector of a line segment, and the bisector of an angle.) 9Gp.01 Use knowledge of bearings and scaling to interpret position on maps and plans.	Pages 348-352** Pages 46-57 Pages 60-71 ** from Mastering Mathematics Book 1
ASSIGNMENT THREE			





Module Four	National Curriculum Link	Cambridge Curriculum Link	Coursebook Pages
Topics: 4.1 Fractions reminder 4.2 Multiplying fractions 4.3 Dividing fractions 4.4 Percentages reminder 4.5 Percentage increase and decrease 4.6 Percentage change	- Use the four operations applied to fractions. - Interpret and use percentages as operators. - Use percentages as operators. - Solve problems involving percentage change.	9Nf.03 Estimate, multiply and divide fractions, interpret division as a multiplicative inverse, (and cancel common factors before multiplying or dividing.) 8Nf.05 Understand percentage increase and decrease, and absolute change.	Pages 110-124 Pages 206-222
ASSIGNMENT FOUR			





Module Five	National Curriculum Link	Cambridge Curriculum Link	Coursebook Pages
Topics: 5.1 Algebra reminder 5.2 Expanding brackets 5.3 Factorising into one bracket 5.4 Rearranging formulae 5.5 Circumference of a circle 5.6 Area of a circle	- Simplify and manipulate algebraic expressions to maintain equivalence. - Substitute numerical values into formulae and expressions. - Understand and use standard mathematical formulae. - Manipulate algebraic expressions by taking out common factors. - Rearrange formulae to change the subject. - Calculate circumference of circles. - Calculate area of circles.	7Ae.03 Understand how to manipulate algebraic expressions including: - collecting like terms - applying the distributive law with a constant. 8Ae.03 Understand how to manipulate algebraic expressions including: - applying the distributive law with a single term (squares and cubes) - identifying the highest common factor to factorise. 8Ae.05 Understand that a situation can be represented either in words or as a formula (mixed operations) and manipulate using knowledge of inverse operations to change the subject of a formula. 8Gg.02 Understand π as the ratio between a circumference and a diameter. Know and use the formula for the circumference of a circle. 9Gg.01 Know and use the formulae for the area and circumference of a circle.	Pages 125-137 Pages 265-281
ASSIGNMENT FIVE			





Module Six	National Curriculum Link	Cambridge Curriculum Link	Coursebook Pages
Topics: 6.1 Equations reminder 6.2 Solving equations with unknown on both sides 6.3 Solving equations with brackets 6.4 Types of quadrilateral 6.5 Area of parallelogram, triangle and trapezium	- Use algebraic methods to solve linear equations. - Apply properties of quadrilaterals to derive results about angles and sides. - Derive and apply formulae to calculate the perimeter and area of triangles. - Calculate and solve problems involving areas of parallelograms and trapezia.	8Ae.06 Understand that a situation can be represented either in words or as an equation. Move between the two representations and solve the equation (integer or fractional coefficients, unknown on either or both sides). 8Gg.01 Identify and describe the hierarchy of quadrilaterals. 7Gg.05 Derive and know the formula for the area of a triangle. Use the formula to calculate the area of triangles and compound shapes made from rectangles and triangles. 8Gg.04 Use knowledge of rectangles, squares and triangles to derive the formulae for the area of parallelograms and trapezia. Use the formulae to calculate the area of parallelograms and trapezia.	Pages 142-161 Pages 162-181
ASSIGNMENT SIX			





Module Seven	National Curriculum Link	Cambridge Curriculum Link	Coursebook Pages
Topics: 7.1 Properties of 3D shapes 7.2 Nets 7.3 Surface area and volume of a cuboid 7.4 Using diagrams to apply Pythagoras's theorem 7.5 Applying Pythagoras's theorem directly	- Use language to analyse 3D shapes. - Use the properties of faces and surfaces to solve problems in 3D. - Apply formulae to calculate volume of cuboids - Use Pythagoras' theorem to solve problems involving right-angled triangles.	7Gg.06 Identify and describe the combination of properties that determine a specific 3D shape. 7Gg.09 Use knowledge of area, and properties of cubes and cuboids to calculate their surface area. 8Gg.08 Use knowledge of area, and properties of cubes, cuboids, (triangular prisms and pyramids) to calculate their surface area. 9Gg.10 Know and use Pythagoras' theorem.	Pages 186-205 Pages 282-295
ASSIGNMENT SEVEN			





Module Eight	National Curriculum Link	Cambridge Curriculum Link	Coursebook Pages
Topics: 8.1 Ratio and proportion reminder 8.2 Conversion graphs 8.3 Best buys 8.4 Finding averages and range from ungrouped frequency tables 8.5 Pie charts	- Divide a given quantity into two parts in a given 'part:part' or 'part:whole' ratio. - Solve problems involving direct proportion and ratio. - Describe and use appropriate measures of central tendency (mean, mode, median). - Construct pie charts.	7Nf.09 Understand and use the unitary method to solve problems involving ratio and direct proportion in a range of contexts. 8Nf.09 Understand and use the relationship between ratio and direct proportion. 7Nf.11 Understand how ratios are used to compare quantities, to divide an amount into a given ratio with two parts. 8Nf.11 Understand how ratios are used to compare quantities, to divide an amount into a given ratio with two or more parts. 7Ss.04 Use knowledge of mode, median, mean and range to describe and summarise large data sets. Choose and explain which one is the most appropriate for the context. 8Ss.04 Use knowledge of mode, median, mean and range to compare two distributions, considering the interrelationship between centrality and spread.	Pages 223-243 Pages 248-264





		8Ss.03 Record, organise and represent categorical, discrete (and continuous) data. Choose and explain which representation to use in a given situation: (- Venn and Carroll diagrams) (- tally charts, frequency tables and two-way tables) (- dual and compound bar charts) - pie charts (- frequency diagrams for continuous data) (- line graphs and time series graphs) (- scatter graphs) (- stem-and-leaf diagrams)	
ASSIGNMENT EIGHT			

